FollowUpBoss Integration

1. Overview

1.1 Description

FollowUpBoss Integration connects **FollowUpBoss** (real estate CRM for agents and teams) with **Newo Platform**.

It enables:

- Creating or updating leads in the CRM with auto-deduplication via `POST /events`
- Looking up existing contacts by phone number to personalize conversations (lead stage, assigned agent)
- Checking appointment availability by querying existing appointments and computing free slots
- Scheduling appointments for showings, consultations, and meetings
- Cancelling existing appointments
- Automatically logging calls and conversation summary notes to contact records after the session ends

This integration is designed for **real estate agents and team administrators** who want to **capture leads and manage appointments through an AI receptionist** (voice or chat).

1.2 Use Cases

- When a caller provides their contact info as a new lead → Create lead via `POST /events` with source tracking
- When a caller is an existing contact → Look up their profile, display lead stage and assigned agent in the conversation
- When a caller asks about available times → Query existing appointments, compute free 30-min slots in 9-17 window, present options
- When a caller books a time → Create appointment in FUB with person ID and assigned agent
- When a caller cancels an appointment → Cancel via `DELETE /appointments/{id}`
- When a session ends → Automatically log the AI call and attach a conversation summary note to the contact

1.3 Supported Features

Feature	Supported	Notes
Lookup Existing Contact	✅	By phone number via `GET /people?phone=`
Create / Update Lead	✅	Via `POST /events` (auto-deduplicates by phone/email)
Check Availability	✅	Computes free slots from existing appointments (9-17 window)
Book Appointment	✅	`POST /appointments` with invitees (person + user)
Cancel Appointment	✅	`DELETE /appointments/{id}`
Post-Call Log	✅	`POST /calls` with outcome, duration, note (fires on session end)

Post-Call Note	✅	`POST /notes` with conversation summary (fires on session end)
Appointment Types Catalog	✅	Cached at setup via `GET /appointmentTypes`
Users / Agents Catalog	✅	Cached at setup via `GET /users`
Auto-Identify by Phone	✅	Fires on `define_user_phone_number` event
Reschedule Appointment	❌	Cancel + rebook as workaround (Phase 2)
Create Follow-Up Task	❌	Planned for Phase 2 (`POST /tasks`)

2. Requirements

Before installation:

- Active **FollowUpBoss** account (trial or paid)
- **FollowUpBoss API Key** (Admin → API)
- **X-System** and **X-System-Key** registered at https://apps.followupboss.com/system-registration (one-time setup per integration)

3. How to Setup

3.1 Step 1 — Obtain FollowUpBoss Credentials

1. Log in to your **FollowUpBoss** account
2. Navigate to **Admin → API** to get your **API Key**
3. Register your integration at https://apps.followupboss.com/system-registration to obtain:
   - **X-System** (system name, e.g., `NewoAI`)
   - **X-System-Key** (system-level API key)
4. Store all credentials securely

3.2 Step 2 — Connect in Platform

1. Open **Newo → Projects**
2. Import `followupboss.zip` to the project
3. Set the following attributes in **Builder / Attributes**:

Attribute	Required	Description
`followupboss_api_key`	✅	FollowUpBoss API key (Admin → API)
`followupboss_x_system`	✅	Registered system name (X-System header)
`followupboss_x_system_key`	✅	Registered system API key (X-System-Key header)
`followupboss_base_url`	❌	API base URL (default: `https://api.followupboss.com/v1`)
`followupboss_default_event_source`	❌	Lead source name (default: `newo.ai`)

followupboss_default_appointment_type_id	❌	Default appointment type ID (auto-populated from catalog)
followupboss_default_agent_user_id	❌	Default agent to assign (auto-populated from users catalog)
followupboss_override_agent_attributes	❌	Override superagent settings on setup (default: True)
followupboss_setup_scenarios	❌	Auto-add canvas scenarios (default: True)
followupboss_enable_post_call_log	❌	Log call to contact timeline after session (default: True)
followupboss_enable_post_call_note	❌	Add conversation note after session (default: True)

4. Click **Save + Publish All**
5. On publish, the SetupFlow automatically:
    - Creates HTTP connector (`followupboss_connector`)
    - Sets superagent booking attributes (`booking_enabled` , `booking_type=project` , `booking_cancellation_enabled` , etc.)
    - Fetches appointment types and users from FUB, caches them
    - Registers 2 custom tools (`followupboss_lookup_contact` , `followupboss_create_lead`)
    - Adds 5 canvas scenarios and 5 intent types (real estate workflows)

4. How to Use

4.1 How the Integration Works

The integration operates based on **Triggers and Actions** via the event-driven system.

- A **Trigger** is a system event that starts a flow
- An **Action** is the API operation performed in FollowUpBoss

Triggers

Trigger	Event	Description
Setup	project_publish_finish	Initializes integration on deploy
Canvas Setup	gm_workflow_builder_canvas_built	Adds custom intents/scenarios to canvas
Auto-Identify Caller	define_user_phone_number	Looks up contact when phone is detected
Lookup Contact Tool	followupboss_lookup_contact	LLM-triggered contact lookup
Create Lead Tool	followupboss_create_lead	LLM-triggered lead creation
Check Availability	get_available_slots	ConvoAgent requests free time slots
Book Appointment	book_slot	ConvoAgent creates appointment

Cancel Appointment	`convo_agent_cancel_booking`	ConvoAgent cancels existing appointment
Post-Call Actions	`custom_session_ended`	Fires on session end → log call + add note

Available Actions

- **Search People** — GET `/people?phone={phone}&limit=1`
- **Create/Update Lead** — POST `/events` (auto-deduplicates)
- **Get Appointment Types** — GET `/appointmentTypes`
- **Get Users** — GET `/users`
- **Create Appointment** — POST `/appointments` with invitees
- **List Appointments** — GET `/appointments?fromDate=...&toDate=...` (for availability)
- **Cancel Appointment** — DELETE `/appointments/{id}`
- **Log Call** — POST `/calls` with personId, outcome, note
- **Add Note** — POST `/notes` with personId, subject, body

4.2 Architecture Diagrams

Overall Sequence

```
sequenceDiagram
    participant U as User
    participant A as ConvoAgent
    participant I as FollowUpBoss Integration
    participant API as FollowUpBoss API

    Note over I: Setup (on publish)
    A->>I: project_publish_finish
    I->>API: GET /appointmentTypes
    API-->>I: Appointment types
    I->>API: GET /users
    API-->>I: Users / Agents
    I->>I: Cache static data + register tools

    Note over U,API: Contact Identified
    U->>A: (phone number detected)
    A->>I: define_user_phone_number
    I->>API: GET /people?phone={phone}
    API-->>I: Contact data (or empty)
    I->>A: ExistingClientDetails updated + urgent_message

    Note over U,API: Availability Check
    U->>A: "What times are available April 10?"
    A->>I: get_available_slots {date, duration}
    I->>API: GET /appointments?fromDate=...&toDate=...
    API-->>I: Existing appointments
    I->>I: Compute 30-min slots in 9-17 range, minus busy
    I->>A: result_success {availability[]}
    A->>U: "Here are available times: 9:00, 9:30, ..."
```

```
    Note over U,API: Booking
    U->>A: "Book 11 AM"
    A->>I: book_slot {personId, appointmentTypeId, dateTime}
    I->>API: POST /appointments
    API-->>I: appointment_id
    I->>A: result_success {booking_id}
    A->>U: "Booking confirmed!"

    Note over U,API: Cancellation
    U->>A: "Cancel my appointment"
    A->>I: convo_agent_cancel_booking {booking_id}
    I->>API: DELETE /appointments/{id}
    API-->>I: 204 No Content
    I->>A: result_success {cancelled}
    A->>U: "Your appointment has been cancelled"

    Note over U,API: Session Ends
    A->>I: custom_session_ended
    I->>API: POST /calls {personId, outcome, note}
    I->>API: POST /notes {personId, subject, body}
```

5. Project Structure

```
followupboss/
├── FollowUpBossIntegration/
│   ├── FollowUpBossIntegration.yaml
│   ├── libraries/
│   │   └── base64/              # encode/decode helpers (Basic Auth)
│   └── agents/
│       └── FollowUpBossAgent/
│           ├── agent.yaml
│           └── flows/
│               ├── SetupFlow/          # Attributes, connectors, tools, static data
load
│               ├── NetworkFlow/        # HTTP abstraction + Basic Auth + X-System
headers
│               ├── ApiFlow/            # Business logic dispatcher (all API calls)
│               ├── AvailabilityFlow/   # get_available_slots → GET /appointments →
compute free
│               ├── BookingFlow/        # book_slot → POST /appointments
│               ├── CancellationFlow/   # convo_agent_cancel_booking → DELETE
/appointments/{id}
│               ├── ExistingClientFlow/ # Auto-identify + manual lookup tool
│               ├── PostCallFlow/       # custom_session_ended → log call + add note
│               ├── CanvasSetupFlow/    # Canvas intents/scenarios
│               └── MetadataFlow/       # Version info
└── akb/
    └── FollowUpBossAgent.yaml
```

6. Authentication

FollowUpBoss uses **two-layer authentication**:

Layer 1 — System Identification (required for integrations)

Every request includes:

- `X-System` : Registered system name
- `X-System-Key` : System API key

Layer 2 — User Authentication (Basic Auth)

HTTP Basic Auth with API key as username, empty password:

```
Authorization: Basic base64(api_key:)
```

The integration includes a **custom base64 library** (`libraries/base64/encode.nsl`) because NSL Jinja2 doesn't expose Python's `base64` module.

7. Canvas Setup

On publish, `CanvasSetupFlow` adds 5 intent types and 5 scenarios to the canvas:

Intent Types

- `followupboss_appointment_booking_intent` — Schedule appointment
- `followupboss_lead_capture_intent` — New lead
- `followupboss_existing_contact_lookup_intent` — Look up existing contact
- `followupboss_property_inquiry_intent` — Property question
- `followupboss_cancellation_intent` — Cancellation request

Scenarios

- `followupboss_appointment_booking_scenario` — Booking flow with code-phrases for availability/booking
- `followupboss_lead_capture_scenario` — Lead capture with "Let me save your information in our CRM now" code-phrase
- `followupboss_existing_contact_lookup_scenario` — Contact identification
- `followupboss_property_inquiry_scenario` — Property Q&A + lead capture fallback
- `followupboss_answering_questions_scenario` — General Q&A
- `followupboss_cancellation_scenario` — Cancellation with "I'm cancelling your booking now" code-phrase

Platform defaults are removed from canvas to avoid conflicts:

- `scenario_sales_appointment_via_calendar` (Google Calendar tool)
- `scenario_home_service_appointment_request`
- `scenario_common_schedule_appointment_agent`
- `intent_type_appointment_scheduling` , etc.

8. Testing

Unit / Integration Tests

```
poetry run pytest test/followupboss/ -v
```

Coverage: **193 tests** (unit + integration) for all flows.

Mock Server

A mock FollowUpBoss API server is available for local testing:

```
make mock-run PRODUCT=followupboss          # Starts server + ngrok tunnel
curl http://localhost:9090/_mock/routes     # List all mock routes
```

Routes defined in `mocks/followupboss/routes.yaml`.

E2E Scenarios

Live end-to-end tests verified on STG:

- `existing_contact_auto_identify` — lookup by phone → persona populated
- `appointment_booking_full_flow` — lookup → availability → book
- `booking_and_cancellation` — existing contact: book → cancel
- `booking_cancellation_new_customer` — new contact: availability → book → cancel
- `new_lead_capture` — new contact → `POST /events` via code-phrase
- `post_call_log_note` — session end → `POST /calls` + `POST /notes`

Scenario protocols in `test/followupboss/end_to_end/`.

9. Rate Limits (FollowUpBoss API)

Sliding 10-second window:

- **Global:** 250 req/10s (with X-System-Key), 125 without
- **POST /events:** unlimited
- **Notes:** 10 req/10s
- **PUT /people:** 25 req/10s

On HTTP 429, `Retry-After` header indicates wait time.

10. Known Limitations

- **Google Calendar hijack** — If customer has `project_attributes_setting_calendar_event_integration=google_integration` enabled, Google Calendar may intercept booking before FUB flow fires. SetupCanvasSkill deletes `scenario_sales_appointment_via_calendar` to prevent this.
- **Appointment reading** — FUB API only returns appointments created IN FUB (not synced from Google Calendar) and belonging to the requesting API user.
- **Session timeout** — Post-call flow fires on `custom_session_ended`, which is platform-driven (`project_attributes_setting_conversation_session_timeout_unfinished_chat`, default 300s).